EXPERIMENT 18

First-Order Predicate Calculus (FOPC) Parser for Logical Expressions

AIM

To implement a First-Order Predicate Calculus (FOPC) parser in Python using NLTK logic expression semantics.

PROCEDURE

1. Import `nltk.sem.logic.LogicParser`.
2. Instantiate the logical expression parser.
3. Parse FOPC formulas containing predicates, quantifiers ($\forall, \exists$), and logical connectives ($\land, \lor, \neg, \to$).
4. Inspect free variables, constants, and predicate structures.
5. Evaluate logical validity and display the structured expression tree.

PROGRAM

```
import nltk
from nltk.sem.logic import LogicParser

parser = LogicParser()

expressions = [
    "all x.(Student(x) -> Studies(x))",
    "exists x.(Dog(x) & Barking(x))",
    "Father(John, Mary) & Mother(Sarah, Mary)",
    "all x.(Human(x) -> mortal(x))"
]

print("--- First-Order Predicate Calculus (FOPC) Parsed Expressions ---")
for expr_str in expressions:
    parsed_expr = parser.parse(expr_str)
    print(f"\nInput Formula: {expr_str}")
    print(f"Parsed Representation: {parsed_expr}")
    print(f"Expression Type: {type(parsed_expr).__name__}")
    print(f"Free Variables: {parsed_expr.free()}")
    print(f"Constants/Predicates: {parsed_expr.constants()}")
```

OUTPUT

```
> python3 experiment_18.py
--- First-Order Predicate Calculus (FOPC) Parsed Expressions ---

Input Formula: all x.(Student(x) -> Studies(x))
Parsed Representation: all x.(Student(x) -> Studies(x))
Expression Type: AllExpression
Free Variables: set()
Constants/Predicates: {Studies, Student}

Input Formula: exists x.(Dog(x) & Barking(x))
Parsed Representation: exists x.(Dog(x) & Barking(x))
Expression Type: ExistsExpression
Free Variables: set()
Constants/Predicates: {Barking, Dog}

Input Formula: Father(John, Mary) & Mother(Sarah, Mary)
Parsed Representation: (Father(John,Mary) & Mother(Sarah,Mary))
Expression Type: AndExpression
Free Variables: set()
Constants/Predicates: {Father, Mary, Sarah, John, Mother}

Input Formula: all x.(Human(x) -> mortal(x))
Parsed Representation: all x.(Human(x) -> mortal(x))
Expression Type: AllExpression
Free Variables: set()
Constants/Predicates: {Human, mortal}
[Process completed with exit code 0]
```

RESULT

Thus, the First-Order Predicate Calculus parser was successfully implemented and utilized to decompose formal logical representations.